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1 May 19, 2026, Tuesday

1.1 Glucose/sucrose solution preparation (Syringe-filtered)

1. Sucrose: 0.24 M syringe filtered (3.29 g in 40 ml DI water)
2. Glucose: 0.26 M(18.70 g in 400 ml for 0.26 M)
3. Glucose: 1.0 M syringe filtered, (7.20 g in 40 ml water for 1 M)

1.2 GUV lipid mixture preparation

DOPC, DOPS, Liss-Rhod, Cholesterol

1.2.1 25% PS GUVs 5% Cholesterol

Content	Concentration	Volume	Mole percent
DOPC	2.5 mg/ml	100 ul	70%
DOPS	2.5 mg/ml	36.9 ul	25%
Cholesterol	2.5 mg/ml	3.5 ul	5%
LissRhod-PE	0.05 mg/ml	59.5	0.5%

1.3 PDMS 7.5%

40 grams of silicone elastomer mixed with 3 grams of curing agent.

1.4 SUV deposit for drying

100 ul of Green SUV was left for drying in MCT

1.5 GUV deposit for drying

15-20 ul of GUV solution prepared today was left for drying on the ITO.

2 May 20, 2026, Wednesday

Make GUV 0% cholesterol solution.

Also, make SUV 5% cholesterol solution

3 May 21, 2026, Thursday

We did electroformation of 5%-Chol,25%-PS,70% DOPC, 0.5% LissRhod-PE. We dipped it in Sucrose 0.24M for 15 minutes before starting electroformation. We cleaned ITOs and spacer with Pure IPA and Pure Ethanol. Then we started electroformation at 2 Vpp sine wave 10 Hz, with the spacer of thickness 1 mm (?). After one hour, we poured 100 ul of GUV solution into 1 ml of Glucose 0.26M We picked 10 ul of that diluted solution and poured it into vessel chamber with 800 ul of media (Glucose 0.26 M) We aspirated with 8 um sized pipette.

- 3.1 We have optimised the fluorescence (0.5%)
- 3.2 We have optimised the yield (Electroformation at 2 volts 10 Hz with spacer).
- 3.3 TODO Measure the thickness of spacer.
- 3.4 TODO Make another spacer with PDMS.
- 3.5 TODO Electroform DOPC GUVs.
- 3.6 TODO Electroform DOPC-PS 50% - Chol 5% GUVs. Make 200 ul of solution
- 3.7 TODO Electroform DOPC-PS 25% - Chol 0% GUVs. Make 200 ul of solution.
- 3.8 TODO Electroform DOPC-F GUVs - Make 100 ul of solution.
- 3.9 How long are the GUVs stable for in time over days?
- 3.10 How to optimise the shape of GUVs (avoid threads and sticking things)
- 3.11 Maximum pressure? (Is bursting pressure independent of size/shape?)
- 3.12 How much time can we aspirate vesicle for below the maximum pressure?
- 3.13 How does the Maximum pressure vary with constituents? (DOPC, DOPC/PS, DOPC/PS/Chol)
- 3.14 Matlab Calculation of Aspiration versus tension.
- 3.15 Image Analysis in Fiji/MATLAB
- 3.16 Matlab calculation of Volume change and area change.

```
import numpy as np
print("Hello,world")
```

```
##### Let's make 90% charged DOTAP-DOPC (Positive, non-fluorescent)
Chemicals = ['DOPS','DOPC','LissRhod-PE', 'Cholesterol', 'NBD-DOTAP']
Concentrations = [2.5,2.5,0.05,2.5,0.01]; # In chloroform
MixtureVolumes = [20,150,0,0,0]; # in microliters
```

```

Molecular_Weights = [809.518,785.593,1300.712,386.65,710.34]

# Number of Weights in mixture
Weights = np.multiply(MixtureVolumes, Concentrations)
# Percentage of moles in mixture
WeightPercentages = np.array(Weights) / np.sum(Weights) * 100

Moles = np.array(Weights)/np.array(Molecular_Weights)
MolePercentages = np.array(Moles) / np.sum(Moles) *100
np.set_printoptions(suppress=True, precision=5)

print('Moles')
print(Moles)
print('Weights')
print(Weights)
# Percentage of moles in mixture
print('weight Percentages')
print(np.round(WeightPercentages,2))
print('Mole Percentages')
print(np.round(MolePercentages,2))

```

4 May 22, 2026, Friday

Analyse

4.0.1 DOPC GUVs 0% Cholesterol

Content	Concentration	Volume	Mole percent
DOPC	2.5 mg/ml	141.2 ul	99.5%
LissRhod-PE	0.05 mg/ml	58.8	0.5%

4.1 20% Cholesterol - DOPC

4.1.1 DOPC GUVs 20% Cholesterol

Content	Concentration	Volume	Mole percent
DOPC	2.5 mg/ml	121.6 ul	79.5%
Cholesterol	2.5 mg/ml	15.1 ul	20%
LissRhod-PE	0.05 mg/ml	63.3	0.5%

4.2 40% Cholesterol - DOPC

4.2.1 DOPC GUVs 40% Cholesterol

Content	Concentration	Volume	Mole percent
DOPC	2.5 mg/ml	98.7 ul	59.5%
Cholesterol	2.5 mg/ml	32.7 ul	40%
LissRhod-PE	0.05 mg/ml	68.7	0.5%

5 May 25, 2026 Monday

We took data for the Zero Pressure Voltage was 9.05 +- 0.05 V. Volume level of the pressure chamber is 13ml. We did pipetting experiments on 100% DOPC GUVs and 60% DOPC-40% Cholesterol GUVs.

6 May 26, 2026, Tuesday

6.0.1 DOPS 25 %, Chol 40%, DOPC 35%

Content	Concentration (mg/ml)	Volume (ul)	Mole percent
DOPC	2.5	56.9	34.5
DOPS	2.5	42.5	25
Cholesterol	2.5	32.4	40
LissRhod-PE	0.05	68.2	0.5

6.0.2 DOPS 50%, Chol 20%, DOPC 30%

Content	Concentration (mg/ml)	Volume (ul)	Mole percent
DOPC	2.5	44.6	29.5
DOPS	2.5	77.9	50
Cholesterol	2.5	14.9	20
LissRhod-PE	0.05	62.5	0.5

7 May 27, 2026, Wednesday

Zero pressure voltage was 7.9 +- 0.05 Volume level was 13 ml for the pressure chamber. We did an experiment for bursting tension and Area-Volume Calculation for 0% Chol-100% DOPC GUVs as well as 40% Chol-100% DOPC GUVs.

8 May 28, 2026, Thursday

Further DOPS mixture preparations We will test 100% DOPS, 50% DOPS-Cholesterol, 50% DOPS - 45% DOPC - 5% Cholesterol

8.0.1 100% DOPS

Content	Concentration (mg/ml)	Volume (ul)	Mole percent
DOPC	2.5	44.6	29.5
LissRhod-PE	0.05	62.5	0.5

8.0.2 50% DOPS, 45%DOPC, 5% Cholesterol

Content	Concentration (mg/ml)	Volume (ul)	Mole percent
DOPC	2.5	73.7	44.5
DOPS	2.5	63.6	50
Cholesterol	2.5	3.5	5
LissRhod-PE	0.05	59.2	0.5

8.0.3 50% DOPS, 50% cholesterol

Content	Concentration (mg/ml)	Volume (ul)	Mole percent
DOPS	2.5	87.6	50
Cholesterol	2.5	41.9	50
LissRhod-PE	0.05	70.5	0.5

9 May 29, 2026, Friday

We analysed data for 100% DOPC GUVs. At a strain rate of 0.005 mN/m/s, we obtain bursting tension of 1 mN/m

10 May 30, 2025, Saturday

We analysed data for 40% Cholesterol-60% DOPC vesicles. We obtain higher bursting tension, at least double.

11 June 1, 2026, Monday

We analyse data from 25%PS-40%Chol GUV pipetting experiments.